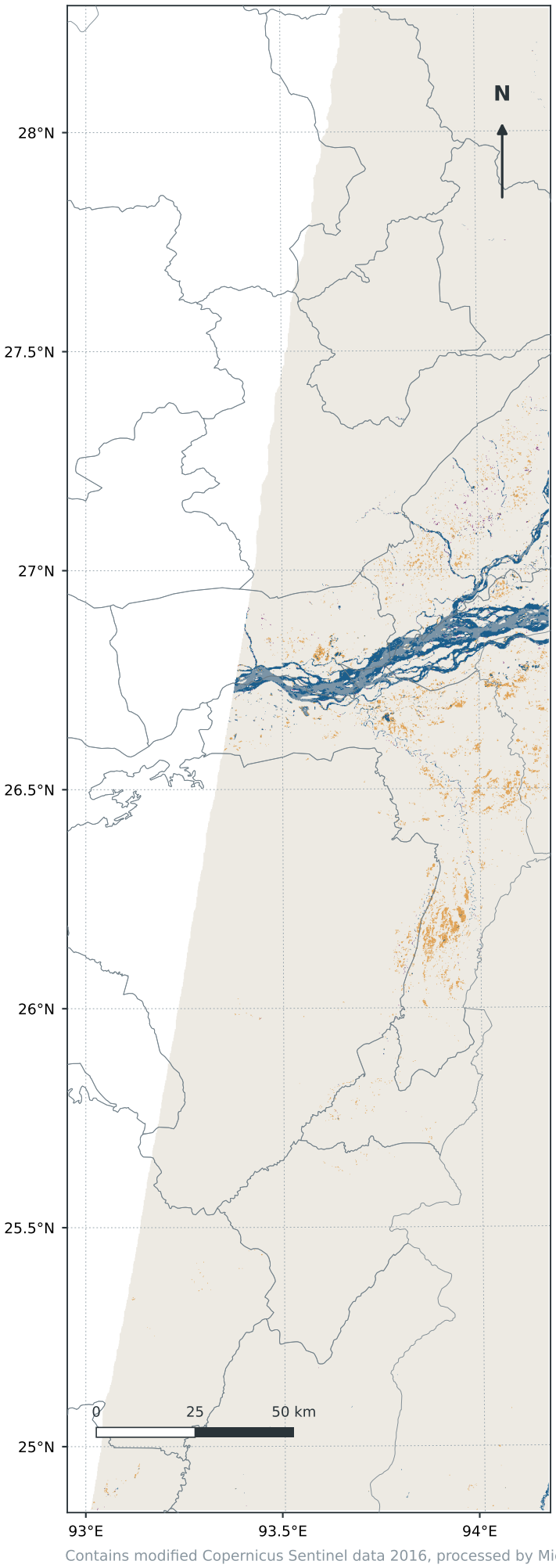
What the flood did between two passes on the same orbit

Sentinel-1A, 7 and 31 August 2016, relative orbit 4, EPSG:32646, 20 m pixel, map-optimal operating point, −18.75 dB



LEGEND

- Drained — *flood on 7 Aug, dry on 31 Aug*
- Still flooded — *flood on both dates*
- Newly flooded — *dry on 7 Aug, flood on 31 Aug*
- Permanent water — *JRC occurrence ≥ 50% on either date*
- Dry throughout
- No data — *not imaged on both dates*

WHAT MOVED, 24 DAYS

class	hectares	share of 7 Aug flood
Flood on 7 Aug	**65,268**	
Drained by 31 Aug	27,933	42.8%
Still flooded	37,334	57.2%
Newly flooded	2,147	3.3%
Flood on 31 Aug	**39,481**	

AGAINST FLOODED CROPLAND

Flooded cropland, 7 Aug	30,109
Flooded cropland, 31 Aug	5,826

Total flood fell 40%.
Flooded cropland fell 81%.

Those rates differ because the water that persists is not sitting on fields. Fields drain within about three weeks. Channels, chars and low wetland do not. Peak extent is the number that gets reported; duration is the number that damages a rice crop.

READ WITH THIS IN MIND

Both dates are relative orbit 4 on the same grid, so every pixel here is compared against itself. That is why this figure uses 7 and 31 August rather than the 12 August peak, which was imaged from a different orbit.

Neither the change classes nor the refinement steps behind them have been validated. The Sen1Floods11 hand labels cover 65 floodplain chips on a single date and contain nothing that speaks to change between two dates. What is drawn here is the difference of two unvalidated maps.

Contains modified Copernicus Sentinel data 2016, processed by Microsoft Planetary Computer. Surface water: JRC/Google. Land cover © ESA WorldCover 2021. Boundaries: geoBoundaries.

Areas were counted at full resolution over the whole scene. The panel is cropped to the ground where something changed and the display is decimated 4×, both for drawing only.